

Apostolos Foivos Varelas



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Profile

As a software engineer, my goal is to connect the theory of machine learning with real-world robotics. I have a strong, practical foundation in Python and sophisticated problem-solving skills because the majority of my recent work has involved satellite data processing and automation. In addition to my professional work, deep learning, RL, and ROS2 have been major topics in my university projects. Right now, I want to use this experience to create clever, practical robotic applications.

Skills

- **Programming**
 - Python
 - C
 - Docker
 - Git
 - Unit Testing
- **Data & Geospatial**
 - NumPy
 - GDAL
 - Rasterio
 - GeoPandas
 - Matplotlib
 - ESA SNAP
- **AI / ML**
 - Neural Networks
 - CNN/RNN/LSTM
 - Transformers
 - LLMs
 - Reinforcement Learning
 - AI Agents
 - PyTorch
- **Robotics & Automation**
 - ROS 2
 - Robotics
 - MuJoCo
 - Robot Navigation

Work Experience

EAGLE PROJECTS

February 2024- Present: Software Developer

- Built remote-sensing pipelines using Sentinel-1/2/3, DEM, ERA5-Land, weather, and AOI data for crop monitoring, risk analysis, subsidence estimation, and irrigation support.
- Developed Sentinel-2 workflows for super-resolution, spectral indices, classification, change detection, and web map tile generation.
- Refactored Sentinel-1 SBAS pipelines for interferogram processing, unwrapping, velocity estimation, DEM creation, and displacement analysis.
- Created agricultural risk and decision models for biomass, flood risk, heat stress, erosion, soil moisture, humidity stress, and irrigation needs.
- Automated geospatial processing with GDAL, Rasterio, xarray, Dask, ESA SNAP, Docker, YAML configs, unit tests, S3 sync, logging, and alerting.
- Produced GIS-ready GeoTIFFs, shapefiles, GeoJSON/NDJSON outputs, PNG quicklooks, PDF reports, summaries, and map tiles.

Education

MASTERS IN AI

University of Piraeus, dep Digital Systems 2024- Present

Thesis: Safe Autonomous Robot Navigation in 3D Settings

BACHELOR IN COMPUTER SCIENCE

University of Piraeus, dep Digital Systems 2020- 2024

Thesis: Reinforcement Learning in a Competitive Environment

ENGLISH CERTIFICATE

C2 Proficiency in English (Michigan ECCE/ECPE)

Personal Projects

Agent-Auction — Multi-Agent Auction Simulator

[Go To Project](#)

- Built a compact multi-agent auction simulator to study bidding strategies and market dynamics across common auction formats.
- Encapsulated auctioneer, bidder agents, and clearing rules; supports reproducible runs and result visualization.
- Tech stack: Python (NumPy, pandas, matplotlib; PyTorch-ready hooks for RL experiments).

DeepFakeDetection — Deep Learning-Based Deep-Fake Video Detection

[Go To Project](#)

- Developed deepfake detectors with ResNet50, Xception+SE, and Swin Transformer; trained/evaluated on FaceForensics++ with strong baselines.
- End-to-end PyTorch pipeline: frame extraction (full-frame & face-crop), mixed precision, learning-rate scheduling, early stopping.
- Metrics & explainability: Accuracy, F1, AUC/PR-AP, EER, MCC; Grad-CAM visualizations at frame and video level.

References available